Vishal Idapalapati
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EDUCATION

TEXAS A&M UNIVERSITY — College Station, TX
MEng in Space Engineering (Focus in AI/ML, Space Robotics/Autonomy & PNT) — Expected: May 2027

- GPA: 4.0

WORK EXPERIENCE

L3 Harris, *Electronic Warfare Systems Engineering Intern – Palm Bay, FL* — June 2024 – September 2024

- Developed and refined MATLAB/Simulink models to enhance satellite tracking and navigation algorithms by 17% for orbital control systems.
- Conducted system level debugging and vibrational testing, optimizing performance reliability by 35% for integrated guidance and control platforms.
- Utilized Git/Bitbucket for version control in development workflows and JIRA/Confluence for system documentation and requirements tracking ensuring traceability and cross functional collaboration.

Northrop Grumman, *Systems Engineering Co-op – Chandler, AZ* — June 2023 – March 2024

- Developed requirement verification models in Cameo as part of MBSE, supporting the full verification lifecycle for XDAM system requirements.
- Collaborated across engineering disciplines and with the customer to develop qualification test plans, complete requirements versus compliance assessments, and confirm verification plans met all customer requirements and system capability needs.
- Authored and edited system requirements and verification documents (SRD, SSS, IRS, ICD) in alignment to mission systems development cycles.
- Produced system architecture diagrams from the SysML modeling to document system design and facilitate cross discipline understanding.
- Refined requirements using DOORS to reflect a fluid system design, ensuring traceability and alignment across evolving technical baselines.
- Conducted reliability analyses, FMECA, and fault tree analysis for mission critical components supporting reliability predictions and control system risk assessments ensuring compliance with technical and mission requirements.

B.H. Aircraft, *Manufacturing Engineering Intern – Ronkonkoma, NY* — May 2022 - August 2022

- Improved cost efficiency by 30% and production time by 65% by redesigning the Trumpf laser layout for key manufacturing templates.
- Operated HAAS CNC machines to produce mode struts and flaps along with the Trumpf laser for cutting, marking, and welding.
- Implemented various improvements along the floor, boosting production time by 20%, and enhancing safety in production.

PROJECTS

U.S. Customs and Border Protection ISR Aircraft (Capstone) — U.S. CBP & Texas A&M University

- Led the avionics sub team with propulsion system design and static thrusting, validating motor and propeller configurations for the UAV.
- Implemented the full avionics architecture and telemetry communication allowing for autonomous flight and secure data transmission.
- Fabricated and assembled the UAV using composite, plywood and 3D printed components to assist with construction. Developed and integrated a dual payload system allowing for a flexible CG.

F-18 Flight Control Systems Design & Simulation — Texas A&M University

- Tuned and designed PID and lead-lag controller in Python for a nonlinear 6-DOF F-18 model with zero steady state error.
- Developed automated analysis scripts for step response, bode plots, root locus and control effort evaluation improving simulation workflow.
- Extracted and analyzed SISO transfer functions from aircraft models, evaluating stability margins, pole zero configurations, and system type.
- Automated system identification and Monte Carlo simulation for control design and producing system performance documentation and tradeoffs.

Geostationary Satellite Mission Design — Texas A&M University

- Optimized a GEO insertion mission from LEO trying to minimize delta v within a seven-day operational window.
- Simulated accurate trajectories using STK Astrogator and modeled Hohmann transfers with plane change maneuvers.

Los Alamos National Laboratory Reconnaissance Glider — LANL & Texas A&M University

- Developed the full avionics architecture integrating flight control, telemetry, and imaging subsystems to meet NIIRS 5 surveillance requirements.
- Designed and modeled the glider folding-wing mechanism using SolidWorks using structural FEA and CFD loads.
- Authored system requirements and conducted risk assessment for different avionics failure modes. Verified through hardware loop simulations.
- Collaborated across aerodynamic, structures, and recovery teams to ensure perfect system integration through all phases of the flight.

CLUB INVOLVEMENT, LEADERSHIP & HONORS

SAE Aero — May 2022 – June 2023

- Performed various management tasks, including fundraising, budgeting, logistical support, and maintaining corporate/sponsor relations.
- Supported team with the manufacturing of the competition plane putting in over a total of 200-man hours over the span of 12 weeks.

Chi Psi Beta Fraternity Inc. President — May 2023 – May 2025

- Trained and led a group of 40+ across various brotherhood, service and fundraising events.
- Partnered with NAMI to bring speakers and members to organized. Raised over $25,000 from various fundraisers for the organization.

Dean's Honor List — May 2022 – Current

Sigma Gamma Tau

SKILLS

- **GNC & Autonomy:** Flight Dynamics, Control Design, Guidance & Navigation, Orbital Mechanics, PNT, Systems M&S
- **Software & Simulation:** Python, MATLAB, C++, Simulink, SysML, Siemens NX, SolidWorks, Git/Bitbucket, Ansys STK
- **Systems Engineering:** Systems Integration, Reliability Predictions & Analyses, Verification & Validation (V&V), Cameo, DOORS
- **Project & Product Management:** Agile Methodology, JIRA, Confluence, Risk Management (FMECA, Fault Tree Analysis)
- **Professional:** Agile, Cross Functional Collaboration, MS Office, Strong Oral and Written Communication, Leadership